MIL-S-11716C
3 April 1972
SUPERSEDING
MIL-S-11716B
13 October 1965

MILITARY SPECIFICATION

SUPERSTRUCTURE, STEEL, FOR BRIDGE, FLOATING, PNEUMATIC

FLOAT, CLASS 60

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the deck panels, filler panels, curbs, ramp panels, and miscellaneous members comprising the superstructure supported on saddled pneumatic floats of a military-type floating bridge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

FF-B-575	- Bolts, Hexagon and Square.
FF-N-836	- Nut: Square, Hexagon, Cap, Slotted, Castellated, Clinch, Knurled and Welding.
FF-W-92	- Washers, Metal, Flat (Plain).
QQ-S-624	- Steel Bar, Alloy, Hot Rolled and Cold Finished, (General Purpose).
QQ-S-781	- Strapping, Steel, Flat and Seals.

FSC 5420

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QQ-T-830	- Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

D2777-2 through -7	- Steel Superstructure.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

A36	- Structural Steel.
A120	- Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless Steel Pipe for Ordinary Uses.
A441	- High-Strength Low Alloy Structural Manganese Vanadium Steel.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

AMERICAN WELDING SOCIETY

A5.1 - Mild Steel Covered Arc-Welding Electrodes.
A5.18 - Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0 - Standard Qualification Procedure.
D2.0 - Welded Highway and Railway Bridges.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The steel superstructure components shall be as shown on Drawings D2777-2 through -7 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e. g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (preproduction model). The supplier shall furnish a superstructure or superstructure component(s), as applicable, for examination within the time frame specified to prove, prior to starting production, that his production methods will produce superstructure components that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Structural steel. Structural steel shall conform to ASTM A36.

3.3.2 Low-alloy structural steel. Low-alloy steel shall conform to ASTM A441.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or A5.18, classification and size as required for the operation.

3.5 Flooring. Flooring shall be 5-inch I-beam, open steel floor, transverse-to-traffic type. Minimum section modulus per 6-inch section of flooring shall be 1.68 inches cubed.

3.6 Bolts, nuts, and washers.

3.6.1 Bolts. Bolts shall conform to FF-B-575, type II, grade 2 steel, UNC-2A.

3.6.2 Nuts. Nuts shall conform to FF-N-836, type II, style 4, grade A steel, UNC-2B.

3.6.3 Washers. Washers shall conform to FF-W-92, type A, grade I, class A, cadmium or zinc plated.

3.7 Connector pins. Connector pins shall be of steel conforming to QQ-S-624, steel number 4140, quenched-and-tempered and cold-finished to produce a minimum yield strength of 125,000 pounds per square inch (psi), minimum ultimate tensile strength of 140,000 psi, and a minimum elongation of 15 percent in 2 inches.

3.8 Steel pipe. Steel pipe shall conform to ASTM A120, black, standard weight, except that the ends shall not be threaded and the hydrostatic test will not be required.

3.9 Steel tubing. Steel tubing shall conform to QQ-T-830, MT 1020 steel, seamless, electric resistance welded or fusion welded type, as hot rolled or as welded condition, round shape.

3.10 Identification marking. Each component shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

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3.11 Treatment and painting. All parts of each superstructure component except as specified herein shall be treated and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Pins, bolts, nuts, washers, and threaded portions of components when applicable shall not be painted but shall be preserved as specified in 5.1.

3.12 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each major structural component of the superstructure such as the deck panels, filler panels, curbs, and ramp panels. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. The markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.13 Workmanship. The fabricated superstructure components shall be free of sharp edges, slivers, or burrs.

3.13.1 Steel fabrication. Steel used in the fabrication of the superstructure components shall be free from laminations, kinks, and sharp bends. Steel shall be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove ash and cooling checks. The straightening of material shall be done by methods that will not cause injury to the metal. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.13.2 Welders and welding.

3.13.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society (B3.0).
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.13.2.2 Welding. The surface of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D2.0 - Welded Highway and Railway Bridges". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

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3.13.2.2.1 Welding jigs. Components shall be assembled in steel jigs or frames and welded while held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characteristics of the metal of the parts being welded.

3.13.3 Interchangeability. Like units shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.13.4 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings. Tolerances for positioning of holes and parts shall be referenced from one end or side of the member and shall not be cumulative.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Steel superstructure components which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance or rejection (see 6.4 and 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction superstructure or component(s), as applicable, shall be examined in accordance with 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each superstructure component shall be inspected for conformance to the dimensions covered by the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings under "Major - 100 percent inspection" Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The superstructure component shall be examined for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. Riveting not as specified.

Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking, incorrect, missing or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials and containers not as specified. Each incorrect material and each incorrect container shall constitute one defect.
108.	Unprotected surfaces not preserved as specified for level A or C.
109.	Bolts, nuts, washers, and pins packaged in quantities exceeding the weight limitation of the box specification for level A.
110.	Components bundled together in quantities other than as specified for level A or B.
111.	Strapping not zinc coated for level A.
112.	Gross weight of wood boxes exceeds 200 pounds for level A.
113.	Marking illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unprotected exterior metal surfaces of components such as pins, bolts, nuts, washers, and threaded portions of components, and any other unprotected metal surface requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Bolts, nuts, washers, and pins. Bolts, nuts, washers, and pins of like description preserved as specified in 5.1.1.1 shall be packaged in close-fitting boxes conforming to PPP-B-636, W6c, in quantities not to exceed the weight limitation of the box specification. The boxes shall be waterproof sealed with tape as specified for slotted style boxes in accordance with the appendix to the box specification.

5.1.2 Level C. The superstructure components shall be preserved and packaged as specified for level A except the boxes for the bolts, nuts, washers, and pins shall be the supplier's commercial fiberboard boxes or boxes conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Loose items. Panel assemblies; panels, deck, filler; panels, ramp, filler; deck units, short; or panels, ramp, tread shall be shipped loose.

5.2.1.2 Bundling. The structural components in the following quantities shall be bundled together: Plate cover (4); plate, cover, filler (2); curb, complete, bridge (3); panel, deck, filler, short (3); curb, short, bridge (5); post, handrail and delineation marker (10); and beam, connector, bridge ramp (2). Strapping conforming to QQ-S-781, type I, heavy duty, class B, grade 2, in appropriate sizes and number of straps per bundle shall be used to secure the items together into bundles.

5.2.1.3 Bolts, nuts, washers, and pins. Bolts, nuts, washers, and pins packaged as specified in 5.1.1.2 shall be packed together in close-fitting boxes conforming to PPP-B-621, class 2, style optional; or PPP-B-601, overseas type, style optional, grade B. The gross weight of each box shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.2.2 Level B. The superstructure components shall be packed as specified for level A except boxes shall conform to PPP-B-640, class 2, style as applicable, and strapping need not be zinc coated.

5.2.3 Level C. The superstructure components shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Loose items, bundled items, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The superstructure assembly is intended for use as the deck section of the class 60 floating bridge.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model (see 3.2).
(c) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

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6.3 Preproduction model. Any changes or deviations of production superstructure components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish superstructure components conforming to this specification.

6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

Custodians:

Army - ME
Navy - MC

Preparing activity:

Army - ME

Project No. 5420-0112

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R288

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-11716C, Superstructure, Steel, For Bridge, Floating, Pneumatic Float, Class 60

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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